DATABASE MANAGEMENT SYSTEM PROJECT SYNOPSIS

Music Streaming Analytics System (Backend-Focused DBMS Project)

Course Name & Code: UCS310 – Database Management Systems
Degree & Year: B.Tech (2nd Year)
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INTRODUCTION

Music streaming platforms such as Spotify and Apple Music generate massive volumes of data from user interactions including song plays, skips, subscriptions, and listening sessions. Every interaction produces valuable data that can be analyzed to understand user behavior and platform performance.

The Music Streaming Analytics System is a backend-oriented DBMS project that focuses on designing, implementing, and analyzing a relational database for music streaming data. The project emphasizes structured data storage, normalization, and analytics using SQL and PL/SQL constructs.

Motivation for Selecting the Problem:
• Real-world data-intensive application
• Strong analytics and reporting use cases
• Hands-on exposure to SQL and PL/SQL
• Understanding backend database design

Importance of DBMS over File-Based Systems:
A DBMS ensures reduced redundancy, data consistency, integrity, concurrency control, and efficient data retrieval, which is not achievable using file-based systems.

PROBLEM STATEMENT

Modern music streaming platforms generate millions of streaming records daily across diverse users, devices, and locations. Managing such large volumes of data while maintaining accuracy, integrity, and fast analytical retrieval is a significant challenge.

File-based or unstructured systems fail to support scalability, complex analytical queries, and data integrity. There is a strong need for a structured relational database that can efficiently manage streaming activity and generate insights using SQL and PL/SQL.

Limitations of Existing Systems:
• Data redundancy and inconsistency
• Poor scalability
• Difficult analytics and reporting
• Lack of integrity constraints

Scope of Database Usage:
The database supports normalized data storage and analytical queries such as most-played songs, top artists, peak listening times, and subscription trends.

OBJECTIVES OF THE PROJECT

• To design a database using ER modeling
• To identify entities, attributes, and relationships
• To convert ER model into relational schema
• To apply normalization up to Third Normal Form (3NF)
• To implement the database using SQL
• To perform analytics using joins and aggregate functions
• To implement PL/SQL procedures, functions, triggers, and cursors
• To ensure ACID properties using transaction management

SCOPE OF THE PROJECT

Functional Boundaries:
• Backend database only
• No frontend development

Types of Users:
• Admin – manages users, songs, artists, subscriptions
• User – streams music
• Analyst – generates reports

Modules Covered:
• User Management
• Song & Artist Management
• Streaming Logs
• Subscription Management
• Analytics Module

PROPOSED SYSTEM DESCRIPTION

The proposed system stores structured streaming data in a relational database. Each streaming activity is logged with user, song, device, and time information. Analytical queries generate insights such as popular songs, top artists, and user listening patterns.

DATABASE DESIGN

Entities:
• Users
• Artists
• Songs
• Subscriptions
• Streams

Relational Schema:
users(user_id PK, name, age, country, created_at)
artists(artist_id PK, artist_name, country, debut_year)
songs(song_id PK, title, artist_id FK, genre, release_year, duration_sec)
subscriptions(subscription_id PK, plan_name, monthly_price, quality)
user_subscription(user_id FK, subscription_id FK, start_date, end_date)
streams(stream_id PK, user_id FK, song_id FK, stream_time, device_type, location, listened_sec)

NORMALIZATION

All tables are normalized through:
• 1NF – removal of repeating groups
• 2NF – removal of partial dependency
• 3NF – removal of transitive dependency

DATABASE IMPLEMENTATION

SQL:
• DDL, DML commands
• Complex SELECT queries
• Views

PL/SQL:
• Procedures
• Functions
• Triggers

• Cursors
• Exception Handling

TRANSACTION MANAGEMENT

COMMIT, ROLLBACK, and SAVEPOINT are used to ensure ACID properties.

TOOLS & TECHNOLOGIES

• MySQL
• SQL
• Stored Procedures & Triggers
• MySQL Workbench

EXPECTED OUTCOMES

• Normalized database
• Efficient analytics
• Automated operations
• Improved data integrity